



CrysDiS

Crystal Diffraction Simulator

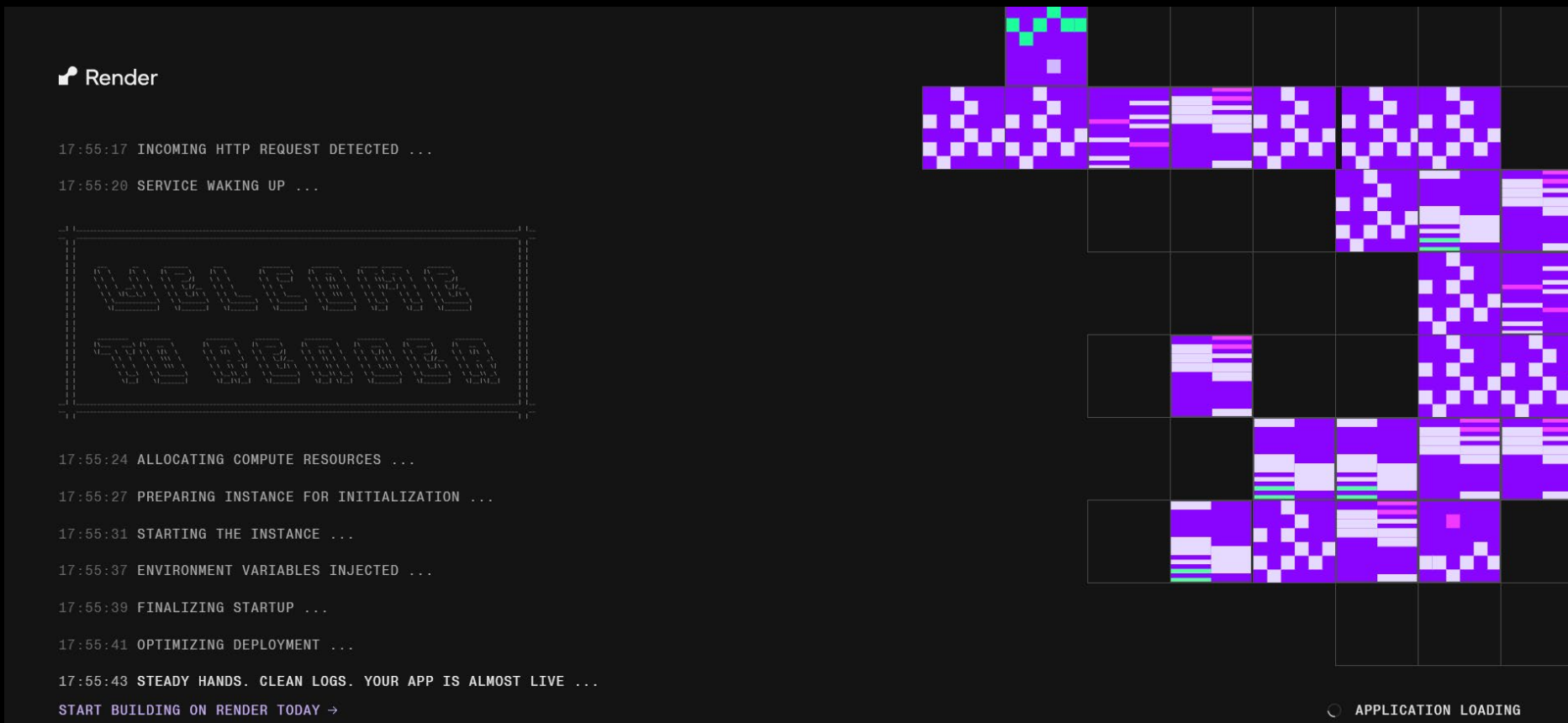
(created using ChatGPT Codex)



Start the program

Option 1: use the website version

- Website: <https://crysdis.onrender.com/>
- Since this is a free URL, it may take 2~3 minutes to warm up
- It also does not have high computational performance
- If you think this software is useful, running it locally is strongly recommended



Start the program

Option 2: download the program and run it locally

- Go to <https://github.com/hepeng1024/CrysDiS> → Click “Releases” → Download the version that matches your OS

The screenshot shows the GitHub repository for CrysDiS. The repository is public and has 0 stars, 0 forks, and 0 watches. The main branch is 'main'. The repository contains several files and folders, including .github/workflows, CrysDiS_tkinter, assets, docs, .dockerignore, .gitignore, CrysDiS.py, Dockerfile, README.md, custom_crystals_localjson, environment.yml, package_macos.sh, package_windows.ps1, pyproject.toml, and requirements.txt. The 'Releases' section is highlighted with a yellow box, showing the latest release v0.1.3, which was released 2 days ago. The 'Assets' section is also visible, showing various download links for different operating systems and architectures.

The screenshot shows the release page for CrysDiS v0.1.3. The release was made by hepeng1024 2 days ago. The release description states: "Add new features to the main CrysDiS: (1) Panel magnification (2) Customized diffraction pattern colors". The 'Assets' section is highlighted with a yellow box, showing the following assets:

Asset	SHA256	Size	Time
CrysDiS-Linux-v0.1.3.tar.gz	sha256:afc29f7e58aef6421e4efad390f8c...	226 MB	2 days ago
CrysDiS-macOS-arm64-v0.1.3.zip	sha256:eb000c0e1e1667e13e1dea42464a1...	115 MB	2 days ago
CrysDiS-Windows-v0.1.3.zip	sha256:efb3b86da80e4b0f1fa30e093...	141 MB	2 days ago
CrysDiS_tkinter-Linux-v0.1.2.tar.gz	sha256:29a0f71d932ea181675547f422890...	193 MB	2 days ago
CrysDiS_tkinter-macOS-arm64-v0.1.2.zip	sha256:1f44981caeee3c4947c8b117e473b...	83 MB	2 days ago
CrysDiS_tkinter-Windows-v0.1.2.zip	sha256:5c0a34cd7721cb37500eb16eb69f1...	111 MB	2 days ago
Source code (zip)			2 days ago
Source code (tar.gz)			2 days ago

The 'Assets' section is also labeled 'Main program'. A green box highlights the Tkinter versions (CrysDiS_tkinter-Linux-v0.1.2.tar.gz, CrysDiS_tkinter-macOS-arm64-v0.1.2.zip, and CrysDiS_tkinter-Windows-v0.1.2.zip). A green text overlay states: "Tkinter versions are also provided if you prefer a more nostalgic interface".

- Please refer to the README file if you need more information

Start the program

Option 3: Run the source code from GitHub

- First install Git and Anaconda/Miniconda. Then clone the repository:

```
git clone https://github.com/hepeng1024/CrysDiS.git
```

```
cd CrysDiS
```

- Create the conda environment:

```
conda env create -f environment.yml
```

```
conda activate crysdis
```

- Start CrysDiS:

```
python CrysDiS.py
```

- Then open this address in a browser:

```
http://127.0.0.1:8080
```

- If port 8080 is already being used, choose another port:

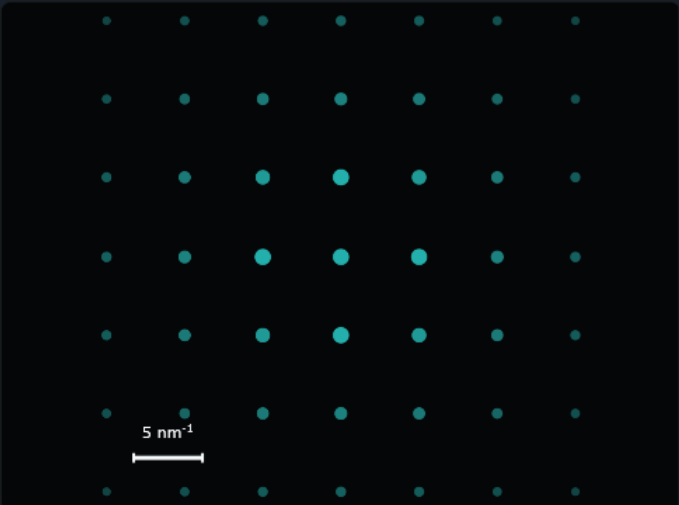
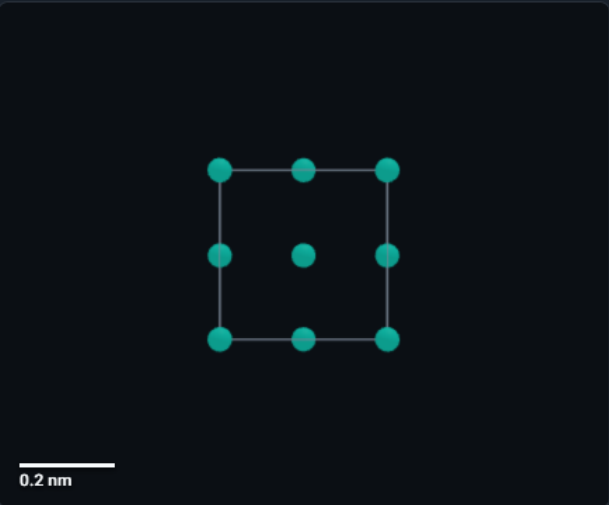
```
NICEGUI_PORT=8094 python CrysDiS.py
```

- Please refer to the README file at <https://github.com/hepeng1024/CrysDiS> for more details

Starting interface

CrysDiS [? INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) [NEW CRYSTAL](#) [ADD COMBO PANEL](#) [ADD PANEL](#)

1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)   



Features of the top toolbar

“ADD PANEL” allows you to have multiple structures at the same time

CrysDIS ? INTRO

☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar **ADVANCED** **CRYSTAL LIST** **LOAD CIF** **NEW CRYSTAL** **ADD COMBO PANEL** **ADD PANEL**

1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: APPLY SYNC

0.2 nm

5 nm⁻¹

2 Crystal: BCC Zone axis: 100 Plane: Vector: Rotation: APPLY SYNC

0.2 nm

5 nm⁻¹

3 Crystal: HCP Zone axis: 100 Plane: Vector: Rotation: APPLY SYNC

0.5 nm

5 nm⁻¹

The image displays the CrysDIS software interface, which is used for visualizing crystal structures and their corresponding diffraction patterns. The interface is organized into three main panels, each representing a different crystal structure: FCC (Face-Centered Cubic), BCC (Body-Centered Cubic), and HCP (Hexagonal Close-Packed). Each panel includes a crystal model, a diffraction pattern, and a scale bar. The top toolbar contains various settings and a highlighted 'ADD PANEL' button, which allows users to add multiple structures simultaneously.

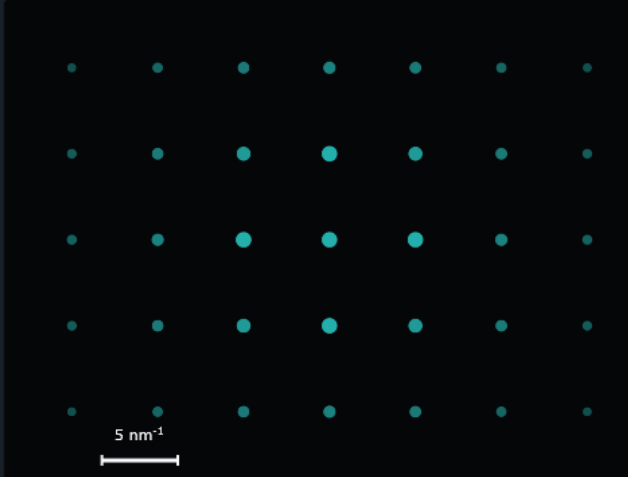
Features of the top toolbar

“ADD COMBO PANEL” allows you to overlap diffraction patterns from different panels and compare them

CrysDis ? INTRO

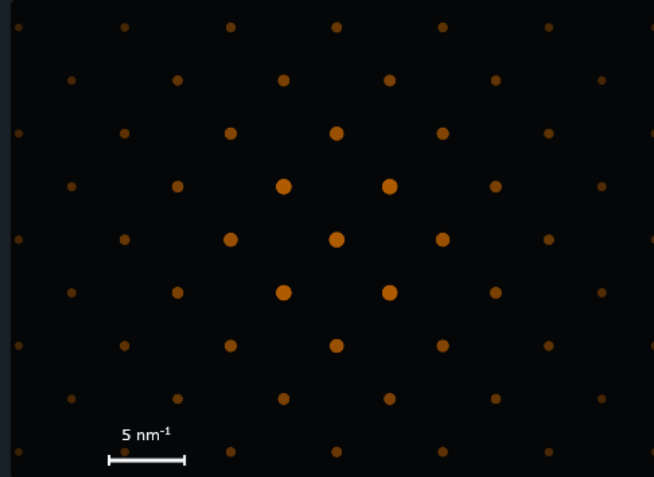
☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar **ADVANCED** **CRYSTAL LIST** **LOAD CIF** **NEW CRYSTAL** **ADD COMBO PANEL** **ADD PANEL**

1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC**



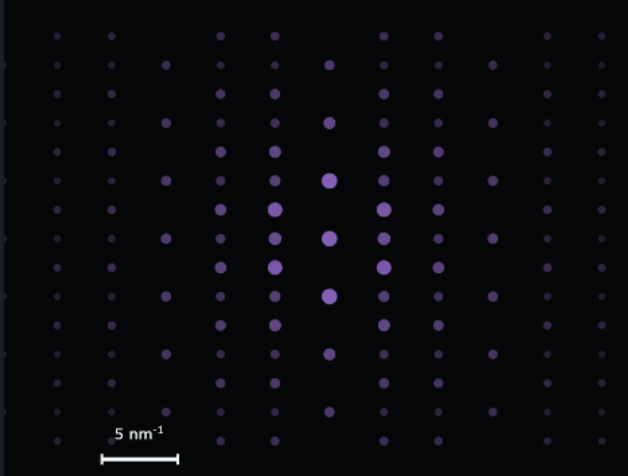
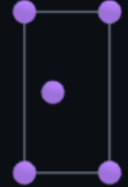
0.2 nm

2 Crystal: BCC Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC**



0.2 nm

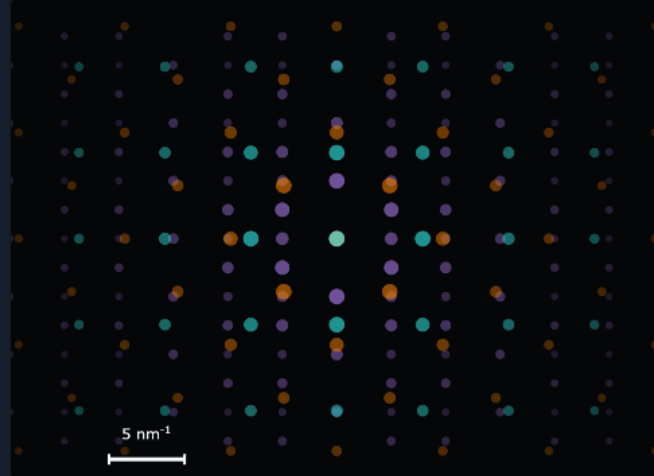
3 Crystal: HCP Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC**



0.5 nm

C1 Current crystals: 1. FCC **ADD** **REMOVE** ☐ Blind crystal motion

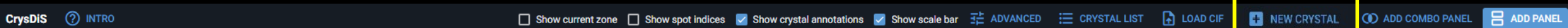
Panel	Crystal	Zone	
1	FCC	[100]	
2	BCC	[100]	
3	HCP	[2-1-10]	



5 nm⁻¹

Features of the top toolbar

“NEW CRYSTAL” lets you build your own crystal



Customized crystal builder

Name Customized crystal	Lattice cubic	Symmetry 1: P1	Save target Project local library	a 0.35 nm
b 0.35 nm	c 0.35 nm	alpha 90 deg	beta 90 deg	gamma 90 deg

APPLY LATTICE CONSTRAINTS

+ ADD ATOM SITE

PREVIEW SYMMETRY

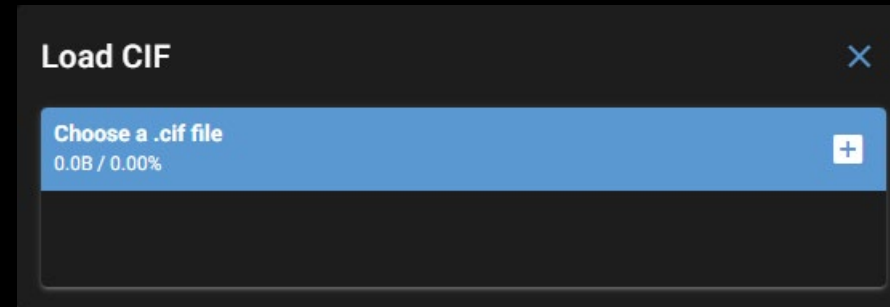
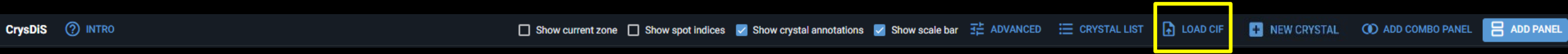
Atom sites

Atom Ni	x 0	y 0	z 0	Occ. 1	Color #10B7A5	Site Ni
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SAVE AS A NEW STRUCTURE

Features of the top toolbar

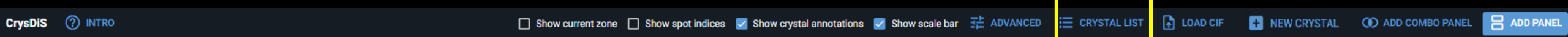
You can load your own .cif files via “LOAD CIF”



Features of the top toolbar

Manage the crystal structures via “CRYSTAL LIST”

Three built-in structures that cannot be edited



Crystal list			✕
	FCC Built-in · cubic, Fm-3m, 1 site(s)		
	BCC Built-in · cubic, Im-3m, 1 site(s)		
	HCP Built-in · hexagonal, P1, 2 site(s)		
	(Al,Ti)Ni3 Custom · cubic, P1, 5 site(s)		
	Al (mp-134) Custom · cubic, Fm-3m, 1 site(s)		
	AlNi3 (mp-2593) Custom · cubic, Pm-3m, 2 site(s)		
	D024 Ni3Ti Custom · hexagonal, P6_3/mmc, 4 site(s)		
	L10 MnNi Custom · tetragonal, P4/mmm, 3 site(s)		
	L12 Ni3Ti Custom · cubic, Pm-3m, 2 site(s)		
	Mg (mp-153) Custom · hexagonal, P6_3/mmc, 2 site(s)		
	Mg5Si6 (mp-568306) Custom · monoclinic, C2/m, 6 site(s)		
	Mg5Si6_no_sym (mp-568306) Custom · triclinic, P1, 11 site(s)		
	Ni3Si (mn-828)		

Features of the top toolbar

Advanced settings in “ADVANCED”

CrysDiS [? INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar **ADVANCED** [CRYSTAL LIST](#) [LOAD CIF](#) [NEW CRYSTAL](#) [ADD COMBO PANEL](#) [ADD PANEL](#)

Advanced settings

Simulation method
Pymatgen TEMCalculator

Thickness
10 nm

Camera length
200 mm

Max hkl
7

Voltage
500 kV

Spot intensity threshold
0.001

Compression factor
100
Contrast of diffraction spots intensity

Spot size scaling
0.3
How large bright spots are

☒ Always snap back view
Check to rotate the crystal back when moving to a new zone

☒ Use four-index basis for hexagonal systems

☐ Auto sync crystal/diffraction

[PALETTE FOR VECTORS/PLANES](#)

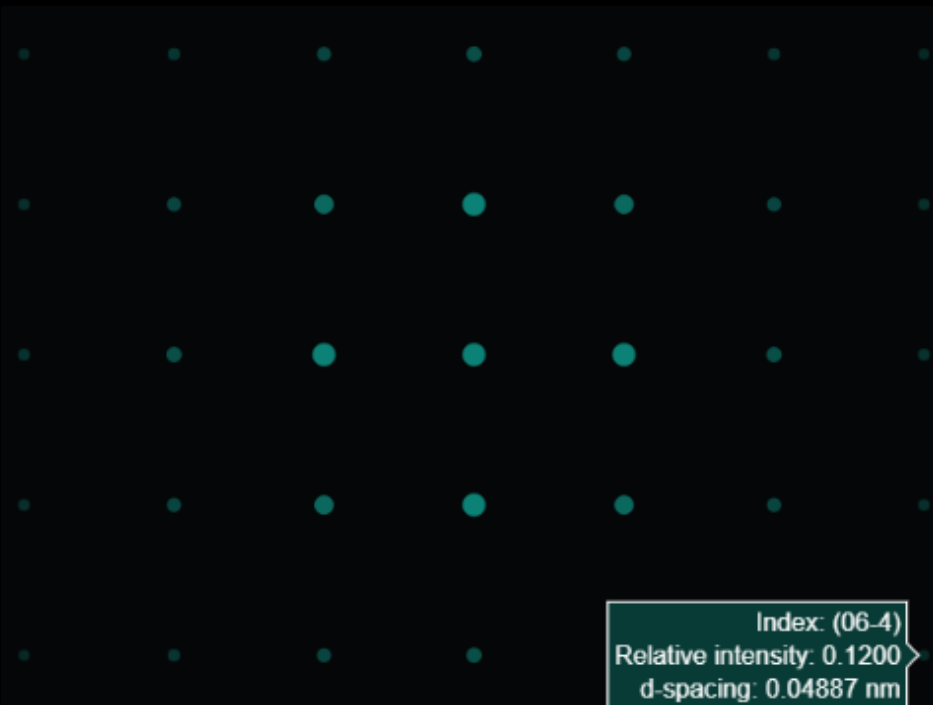
[DIFFRACTION PATTERN COLORS](#)

Log
Ready
Added a comparison panel
Added a comparison panel
Added a combo panel

More about advanced settings

- Compression factor

Compression factor = 10



Compression factor = 150



Dim spots are more visible

More about advanced settings

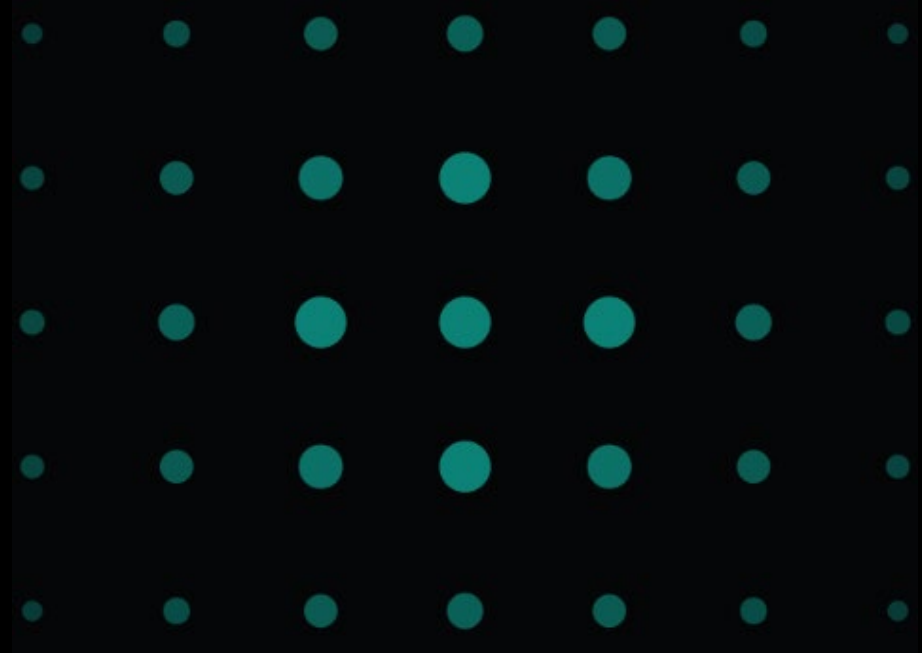
- Spot size scaling effect

Scaling factor = 0



All spots are the same size

Scaling factor = 1



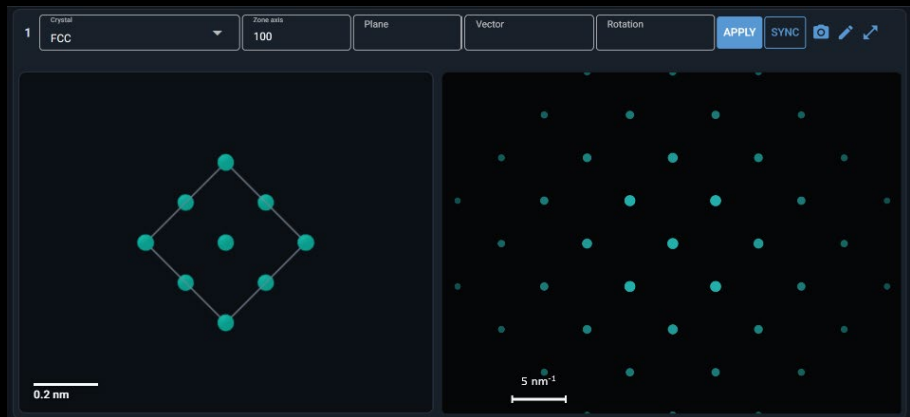
Spot size scales significantly with intensity

More about advanced settings

- Always snap back view



In-plane rotation by 45 deg

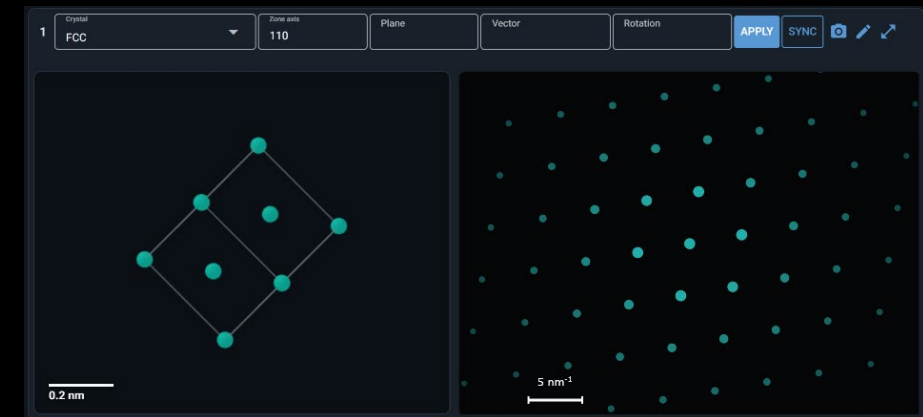


Go to the 110 zone

“Always snap back view” is checked



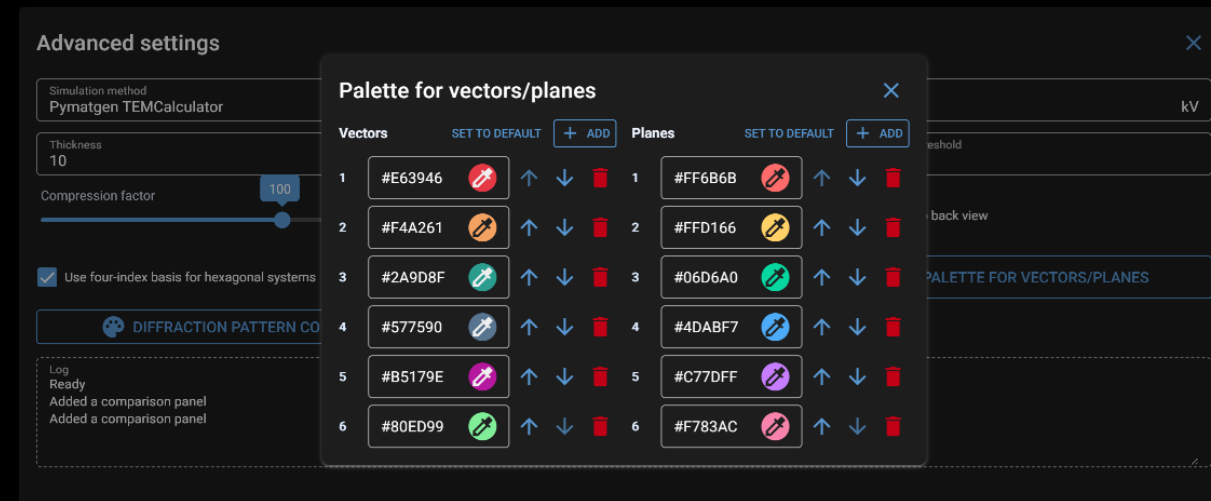
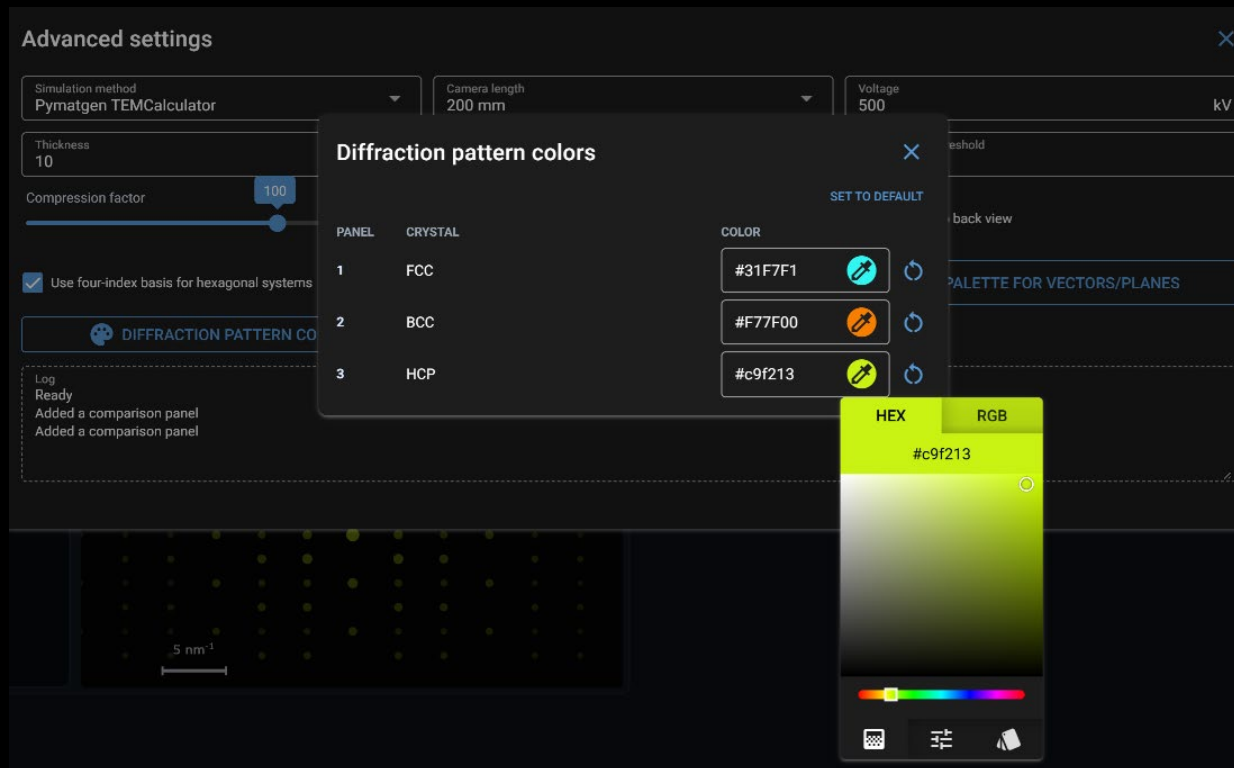
“Always snap back view” is unchecked



The in-plane 45 deg rotation history is retained

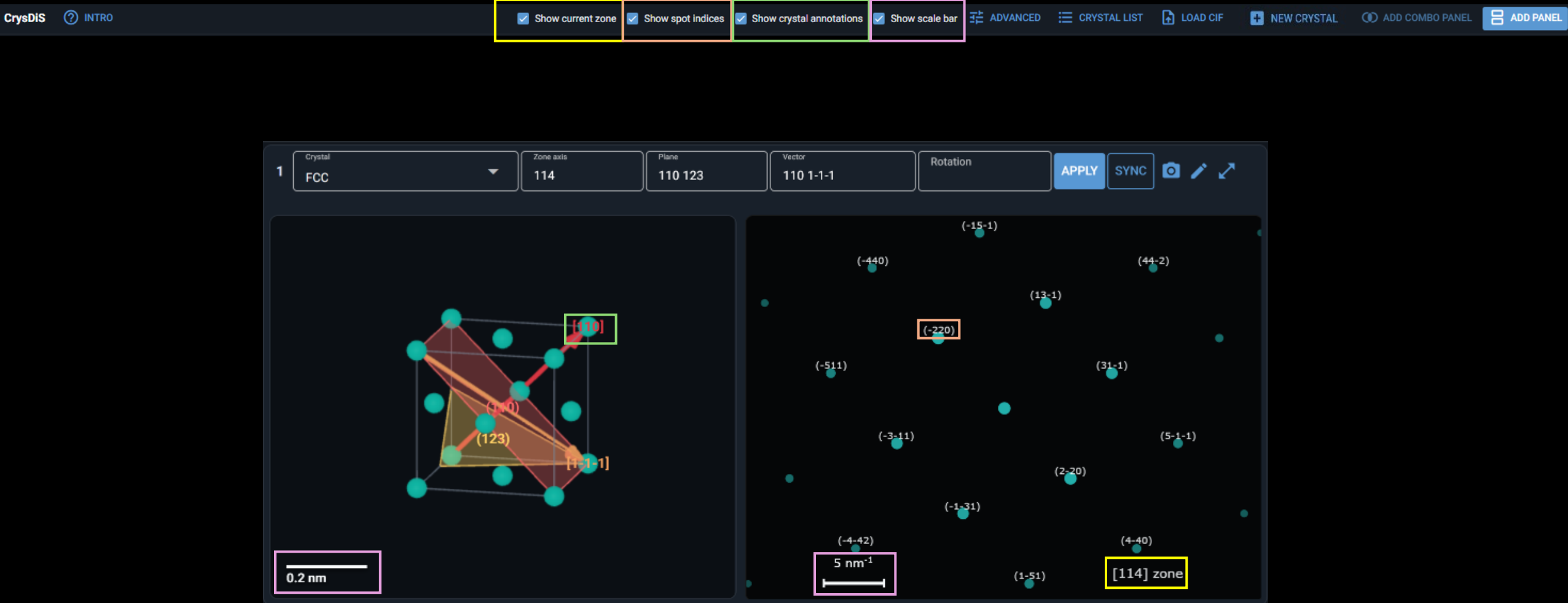
More about advanced settings

- You can also set the colors of diffraction patterns and crystal vectors/planes



Features of the top toolbar

Some annotation toggles



Features of the panel toolbar

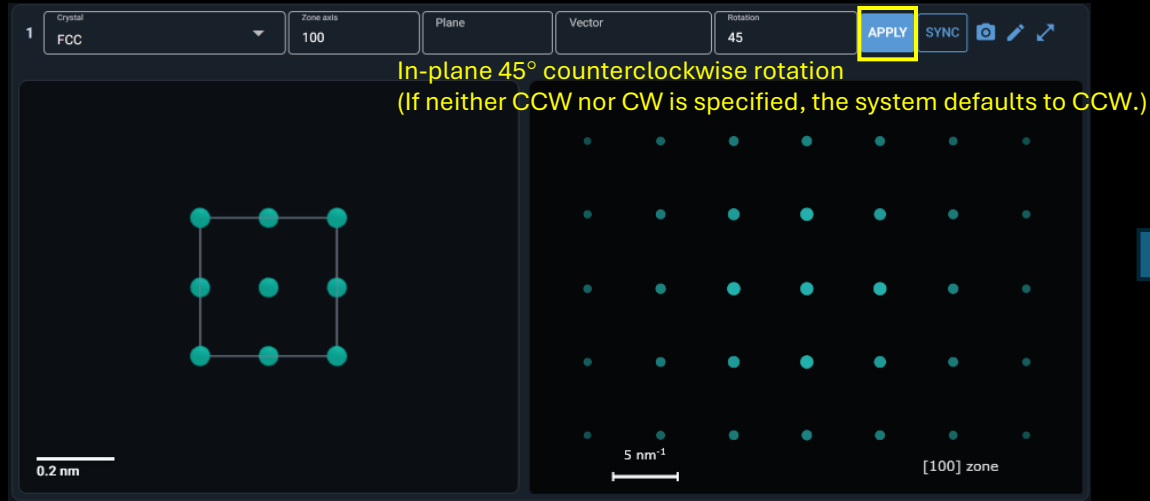
- Choose any crystals in the list or upload/build a new one
- You can go to any zone axis, create vectors/planes
- Reciprocal lattice vectors can be created using *hkl (ex. *110, *0-10)
- For hexagonal systems, 4-index basis inputs are allowed (ex. 2-1-10, *0001)



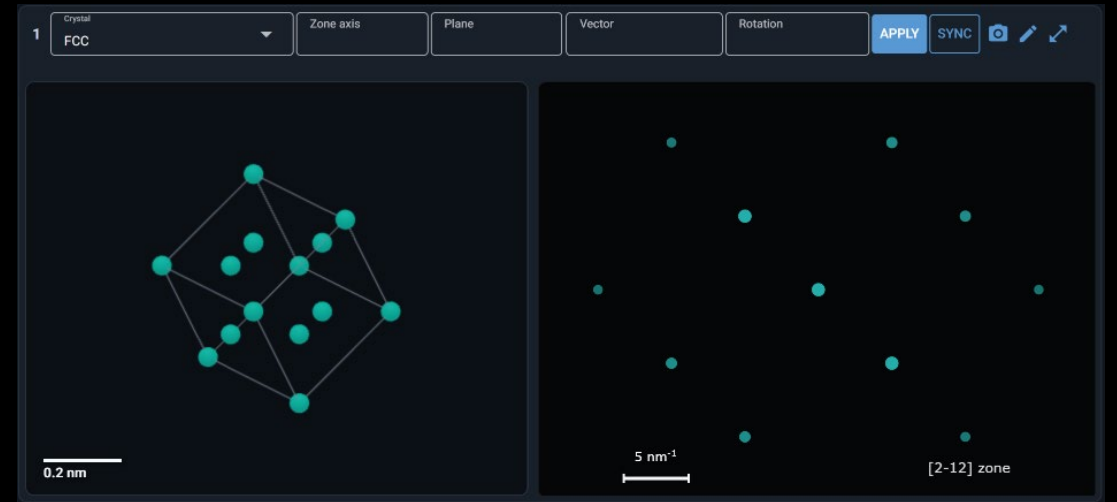
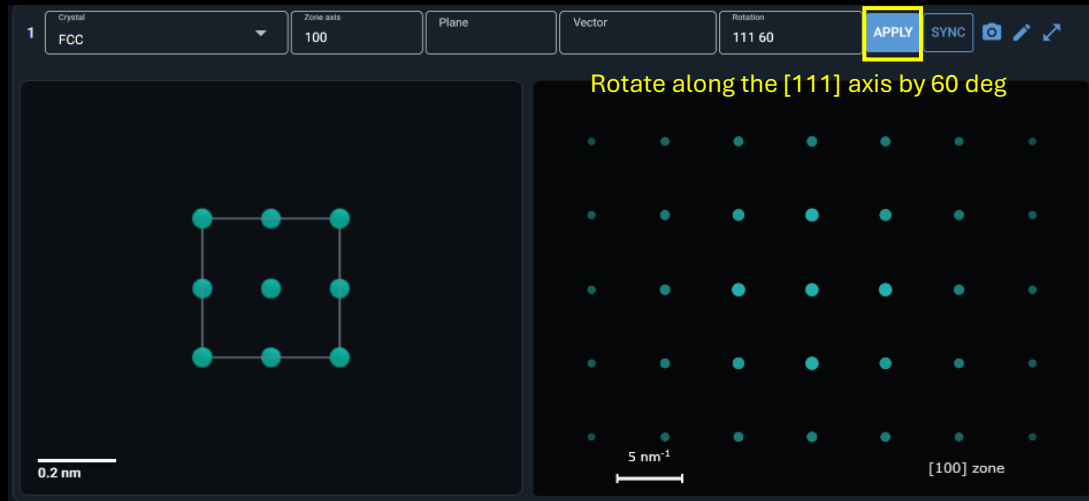
Features of the panel toolbar

- You can also rotate the crystal along a certain axis (right-hand rule is applied)

In-plane rotation: use “deg”, “deg cw”, or “deg ccw”

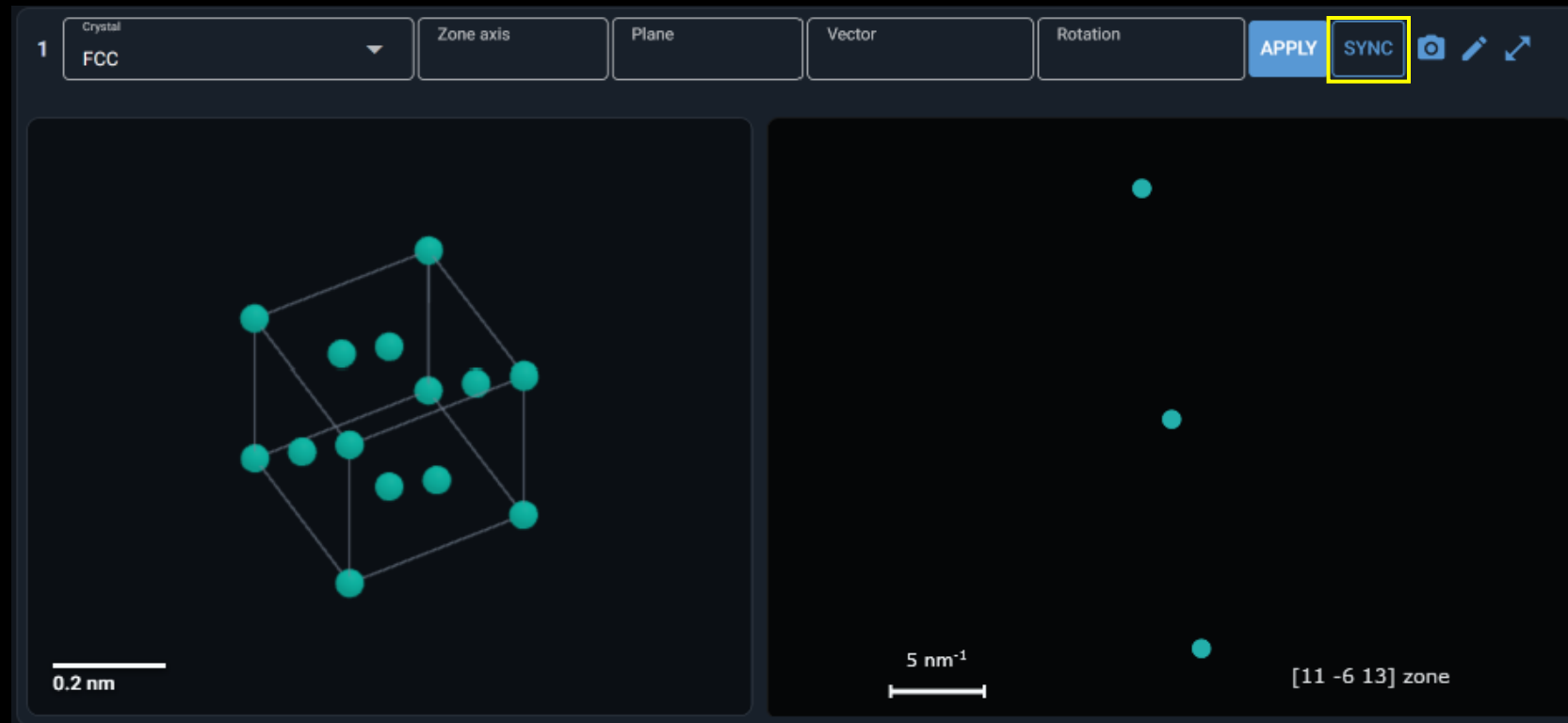


Rotation along a certain axis: use “vector deg”



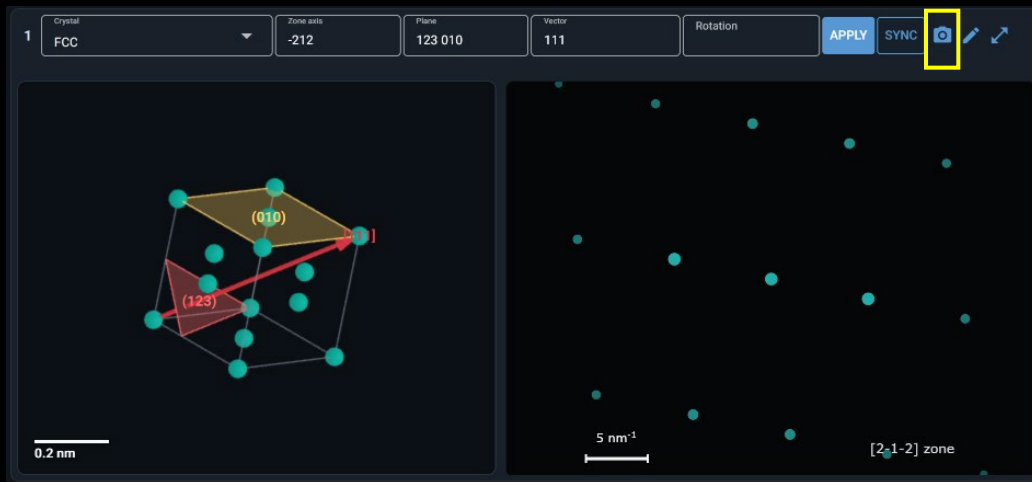
Features of the panel toolbar

- Synchronize the crystal and diffraction pattern using “SYNC”
- It will detect the closest zone axis
- You can also turn on the “Auto sync crystal/diffraction” in the advanced settings



Features of the panel toolbar

- Download the crystal/diffraction pattern at the current viewing angle



Download panel 1 ✕

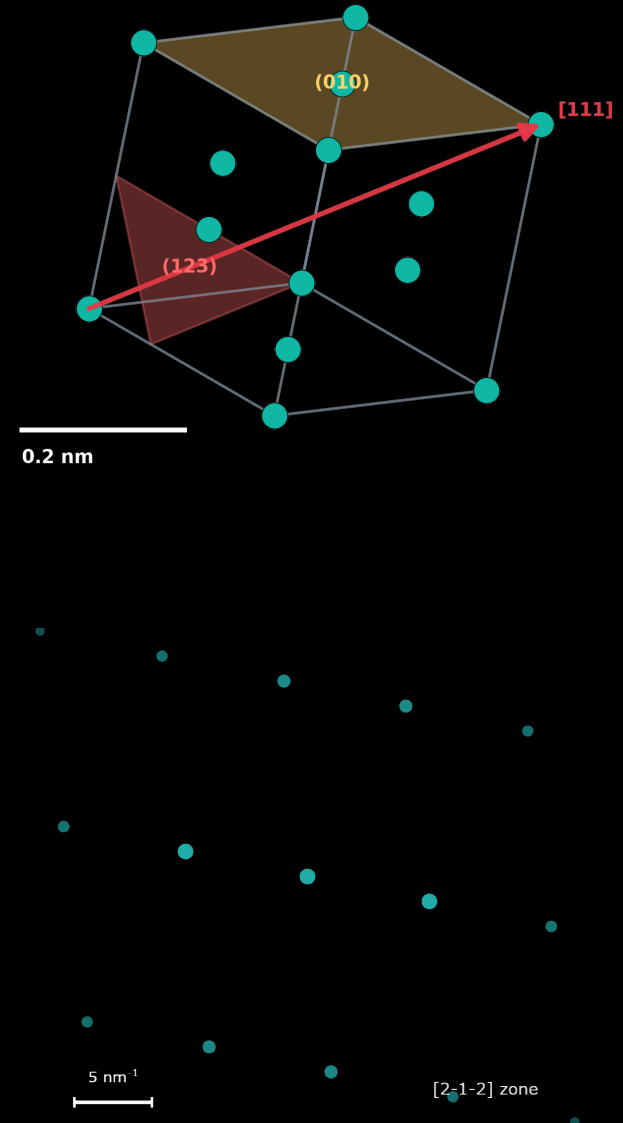
☒ Real-space crystal
☒ Diffraction pattern
☒ Transparent background

Quality
300 dpi

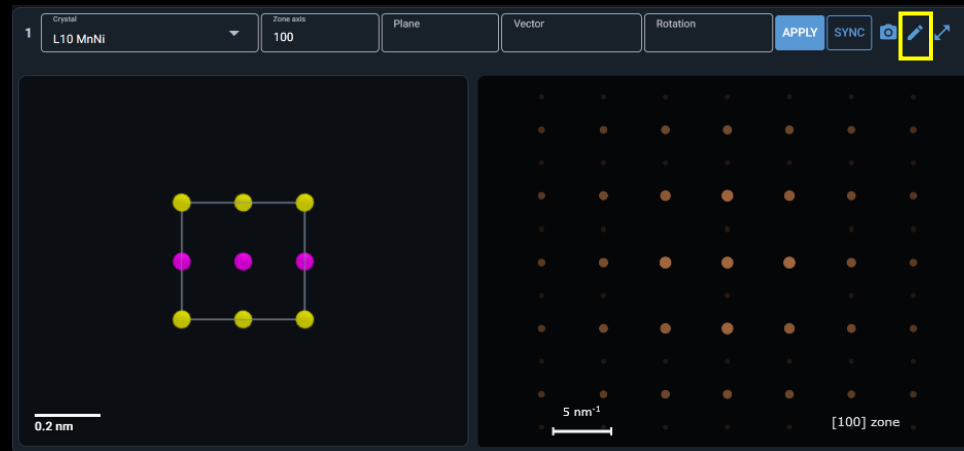
Export folder: C:\Users\haoen\Downloads

📁 CHOOSE EXPORT FOLDER

CANCEL 📄 DOWNLOAD



Features of the panel toolbar



- Edit the current structure

Customized crystal builder

Name L10 MnNi	Lattice tetragonal	Symmetry 123: P4/mmm	Save target User app-data library	a 0.374 nm
b 0.374 nm	c 0.352 nm	alpha 90 deg	beta 90 deg	gamma 90 deg

 APPLY LATTICE CONSTRAINTS  ADD ATOM SITE  PREVIEW SYMMETRY

Atom sites

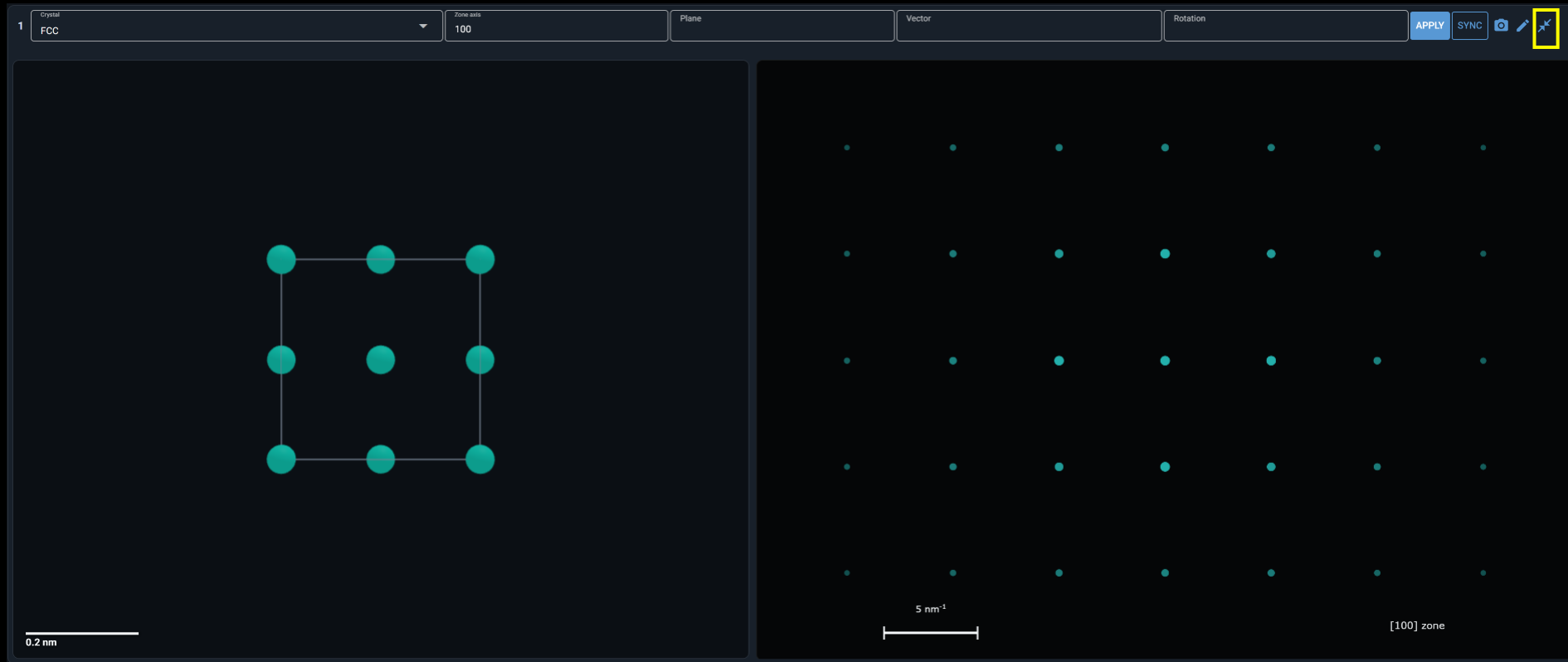
Atom Mn	x 0	y 0	z 0	Occ. 1	Color #dede00		Site Mn1	
Atom Mn	x 0.5	y 0.5	z 0	Occ. 1	Color #dede00		Site Mn2	
Atom Ni	x 0.5	y 0	z 0.5	Occ. 1	Color #ed02e9		Site Ni1	

 SAVE EDITED STRUCTURE  SAVE AS A NEW STRUCTURE

Features of the panel toolbar



- Magnify the panel



- Shrink the panel back

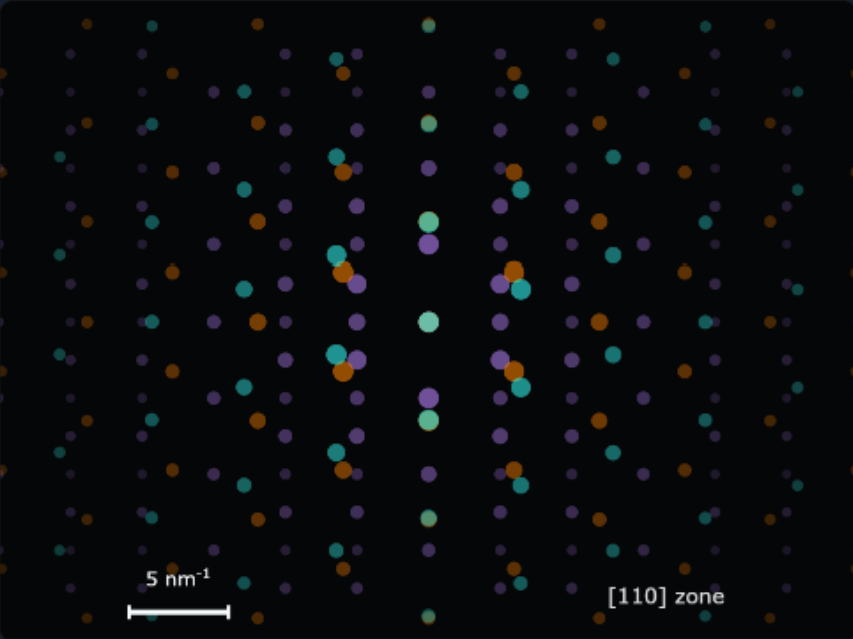
Features of the combo panel

Select crystals present on the page

Add or remove it

C1 Current crystals 1. FCC ADD REMOVE ☐ Bind crystal motion   

Panel	Crystal	Zone	
1	FCC	[110]	  
2	BCC	[111]	  
3	HCP	[2-1-10]	  



Ex. FCC diffraction pattern will be shown on top

The current zone shows the zone axis of the crystal on the top-most layer

The order of the crystal in the list determines the order of the diffraction patterns

Features of the combo panel

Check “Bind crystal motion” if you want the crystals to synchronize their motions

C1

Current crystals
1. FCC

ADDREMOVE☒ Bind crystal motion

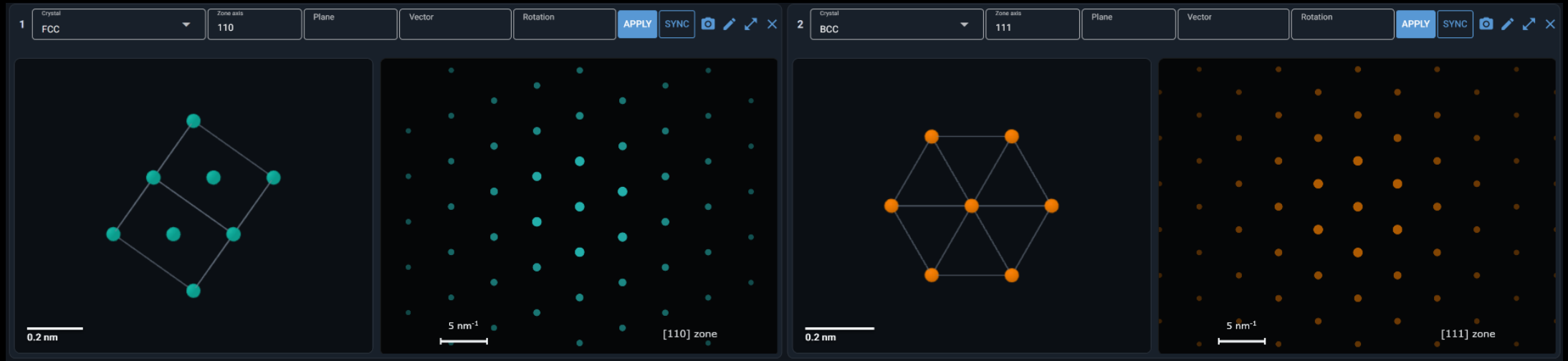
Panel	Crystal	Zone	
1	FCC	[110]	↑ ↓ ×
2	BCC	[111]	↑ ↓ ×

5 nm⁻¹

[110] zone

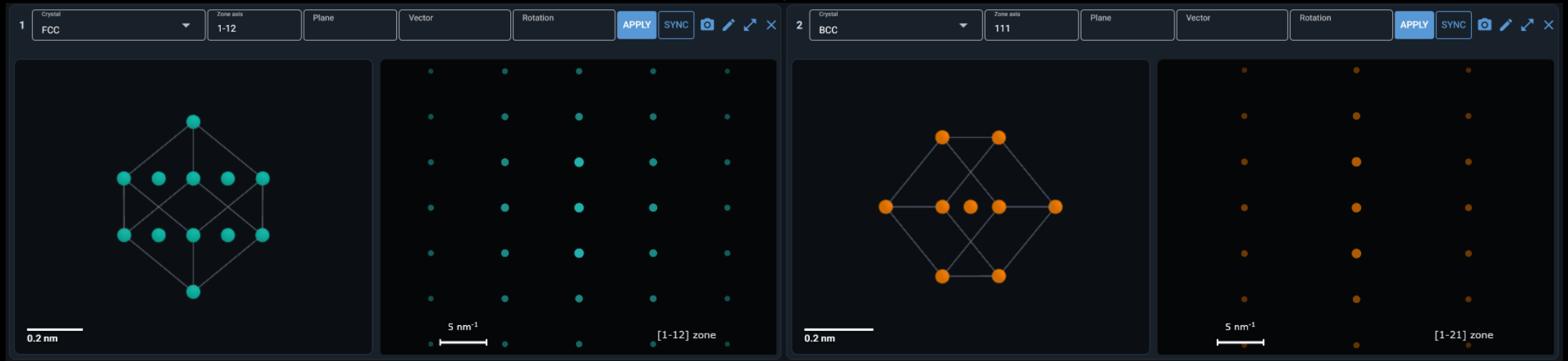
Features of the combo panel

- Bind crystal motion



Ex. if you go to FCC 1-12 zone in panel 1

BCC in panel 2 will do the same motion, and the resulting zone is 1-21



In this case, it means crystals 1 & 2 have the orientation relationship: $[110]_{\text{FCC}} // [111]_{\text{BCC}}$

Comparison of two simulation methods

Simulation method	Ewald Sphere Kinematic	Pymatgen TEMCalculator
Main purpose	More TEM-like kinematic spot visibility	Fast crystallographic zone-axis pattern
Uses voltage?	Yes, through electron wavelength and Ewald sphere radius	Only for wavelength calculation
Uses thickness?	Yes	No
Includes relrod / excitation error?	Yes	No
Includes Ewald sphere curvature?	Yes	No
Good for	Showing why spots appear/disappear with tilt/thickness	Clean ideal zone-axis indexing/intensity